

TORSIONVERSE SERIES 2: PIEZOELECTRICITY AND THE ELECTRON ESCALATOR

Author: Robert Jobson | Date: 2026-08-26

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Companion script: analysis/demos/acoustics_doc.py -- 27/27 PASS [AC1-AC15, SP1-SP7c]

THESIS

Three apparently unrelated phenomena -- piezoelectricity, acoustic amplification at specific geometric apex angles, and apparent weight reduction in spinning pentagonal structures -- are all manifestations of T_{1g} pressure-mode coupling in the icosahedral elastic medium. The coupling efficiency in each case is determined by $\chi(T_{1g})$ from the I_h character table. The unifying quantity is proximity to Maxwell-critical icosahedral geometry ($3V-E=6$).

ABSTRACT

We derive a unified mechanism for piezoelectricity and geometric charge concentration from the torsionverse I_h elastic medium framework. Applied stress tunes a crystal toward the Maxwell-critical ($3V-E=6$) icosahedral configuration, activating T_{1g} mode coupling and generating a voltage. The same coupling, evaluated geometrically for a pentagonal pyramid with face slope $\arctan(1/\phi) = 31.72^\circ$, predicts maximum charge concentration at the apex via a cell-compression channeling mechanism -- the "electron escalator." The 5-sided pyramid achieves $\chi(T_{1g}, C5) = \phi = 1.618$; the 4-sided pyramid gives $\chi = -1$ (defocusing); the 3-sided gives $\chi = 0$ (null). The predicted apex charge ratio 5-sided / 4-sided = ϕ is falsifiable at bench scale with an electrometer and three machined pyramids for under \$50. We identify retrodictions of existing acoustic and crystal data and describe the discriminating experimental protocol.

SECTION 1: INTRODUCTION

Three apparently unrelated phenomena -- piezoelectricity in quartz crystals, acoustic amplification at specific geometric apex angles, and charge concentration at pyramid tips -- are each explained by the same underlying quantity: $\chi(T_{1g})$ from the I_h icosahedral character table. In the torsionverse framework the elastic medium has I_h symmetry, and the T_{1g}

irreducible representation carries the pressure/EM mode. Any physical geometry that activates T_{1g} coupling will produce an observable electric or acoustic response; the coupling strength is given exactly by $\chi(T_{1g})$ evaluated at the rotation angle that characterises the geometry.

This paper derives the mechanism for two cases: (1) piezoelectricity, where applied stress drives a crystal toward the Maxwell-critical icosahedral configuration and activates T_{1g} coupling; and (2) the electron escalator, where a pentagonal pyramid with face slope $\arctan(1/\phi) = 31.72^\circ$ presents exactly the icosahedral face orientation on all five faces simultaneously, causing coherent cell compression along every face and channeling charge directionally toward the apex. Both mechanisms are governed by the same group-theoretic number, $\phi = \chi(T_{1g}, C_5) = 1.618$, and both produce specific falsifiable predictions at bench scale.

The dynamic extension -- apparent weight reduction in a spinning pentagonal pyramid -- is covered in the companion paper doc_series2_spinning_pyramid.txt.

SECTION 2: MAXWELL CRITICALITY AND T_{1g} ACTIVATION

The icosahedral Jobson cell satisfies the Maxwell stability criterion $3V-E=6$ exactly: it has 12 vertices and 30 edges, giving $3(12)-30 = 6$. A truss satisfying $3V-E=6$ sits at the rigidity transition -- it is neither floppy (underconstrained) nor over-rigid. At this critical point the medium supports a soft mode: the T_{1g} pressure mode can propagate through it with zero energy cost in the long-wavelength limit. Below criticality T_{1g} is gapped (suppressed); above criticality the medium is rigid and T_{1g} cannot propagate freely. Exactly at criticality: T_{1g} activates and any deformation that has T_{1g} character produces an electric field response.

The coupling strength is $\chi(T_{1g}, C_N)$ where C_N is the rotation symmetry of the deformation:

$\chi(T_{1g}, C_5)$	= ϕ	= 1.618	[72 deg, icosahedral pentagonal axis]
$\chi(T_{1g}, C_4)$	= -1		[90 deg]
$\chi(T_{1g}, C_3)$	= 0		[120 deg, exact null]
$\chi(T_{1g}, C_2)$	= -1		[180 deg]

The sign matters: positive χ means the T_{1g} mode is constructively activated (pressure focusing, charge directed inward/upward); negative χ means destructive coupling (charge scattered); zero means no coupling at all.

[AC1-AC7, 14/14 PASS]

SECTION 3: PIEZOELECTRICITY AS MAXWELL-CRITICAL TUNING

3.1 MECHANISM

Crystalline quartz (SiO₄ tetrahedra in a helical arrangement) has approximate

icosahedral local symmetry at each Si-O4 tetrahedral cluster. In the unstressed state, the local geometry sits slightly below Maxwell criticality: T_{1g} coupling is weak and no spontaneous electric field appears. When mechanical stress is applied along the crystallographic c-axis, the SiO4 clusters deform toward the Maxwell-critical icosahedral configuration. As they approach criticality, T_{1g} coupling activates and an electric field develops: the piezoelectric effect.

The generated voltage is proportional to:

$$V \propto (\text{proximity to Maxwell criticality}) \times \chi(T_{1g}, C_N)$$

where C_N is the rotation symmetry of the stress-deformed local geometry. For stress along the quartz c-axis (which has approximate 3-fold symmetry):

$$\chi(T_{1g}, C_3) = 0 \rightarrow \text{no piezo response along c-axis}$$

For stress along the a-axis (2-fold symmetry):

$$\chi(T_{1g}, C_2) = -1 \rightarrow \text{piezo response, with polarity depending on sign of the deformation toward vs away from Maxwell criticality.}$$

3.2 PREDICTIONS

- (a) Optimal piezo material: crystal whose local geometry sits closest to Maxwell criticality at rest, requiring the least stress for activation. Prediction: quasicrystals (Al-Mn, Icosahedrite) are AT exact Maxwell criticality throughout. They are not piezoelectric by the standard argument (lack of polar axis), but they should exhibit anomalous T_{1g} coupling at specific resonance frequencies -- measurable as anomalous EM response under vibration, distinct from conventional piezoelectricity.
- (b) Optimal stress axis: the axis whose deformation has the HIGHEST $|\chi(T_{1g}, C_N)|$. For quartz this predicts the a-axis > c-axis, which is consistent with measured quartz piezo coefficients ($d_{11} > d_{33}$).
- (c) Chiral icosahedral cluster: a material with a permanently chiral icosahedral local structure would have permanent T_{1g} coupling with no applied stress -- a pyroelectric-like effect with no temperature dependence. No such material is currently known; this is a design target.
- (d) Curie temperature behavior: thermal fluctuations randomize the local geometry, degrading the Maxwell-critical configuration. The piezo coefficient should decrease with temperature as the local order degrades, vanishing at the Curie temperature when thermal disorder exceeds the Maxwell-criticality well depth.

3.3 RETRODICTIONS

The standard quartz piezo coefficient $d_{11} \sim 2 \text{ pC/N}$ is consistent with a near-but-not-at-Maxwell-critical local geometry where a small finite stress is needed to reach criticality. The framework does not currently derive the numerical value of d_{11} from first principles (the gap from current geometry to Maxwell criticality in unstressed quartz is not yet parameterised); this is an open quantitative item.

SECTION 4: THE ELECTRON ESCALATOR — PENTAGONAL PYRAMID GEOMETRY

4.1 MECHANISM

The optimal charge-concentrating pyramid has face slope $\arctan(1/\phi) = 31.72^\circ$

deg. This slope places each of the five triangular faces at exactly the icosahedral face orientation — the same orientation as the top-cap faces of a Jobson cell viewed along its C5 axis. [AC12, PASS]

Every electron arriving at any point on a given face hits Jobson cells at this icosahedral orientation. With five faces all at the same orientation (rotated 72° from each other around the apex axis), ALL face-area electrons contribute coherently to the same compression pattern in the cells beneath each face. The same mechanism operates along the edges: electrons from all five edges strike the same cell sets along each edge. Both faces and edges drive the same directed compression.

The compressed cells block lateral exit paths. When a new electron enters the region it cannot push laterally — the compressed cells already occupy those directions. The only remaining path is toward the apex, exactly as an electron follows a proton through the medium. The geometry forces directionality; charge accumulates at the apex as a consequence, not as a cause.

Because the face-area contribution dominates (face area $\gg$ edge line area), the critical manufacturing tolerance is face angle precision — hold $\arctan(1/\phi) = 31.72^\circ$ to $\pm 0.5^\circ$ across the face. Edge straightness is a secondary concern.

4.2 N-SIDED CHI TABLE AND PREDICTED CHARGE RATIOS

The T_1g coupling for an N-sided pyramid depends on the rotation angle between opposite faces at the apex:

3-sided pyramid: $\chi(T_{1g}, C3) = 0$	→ no coupling, no escalator (null)
4-sided pyramid: $\chi(T_{1g}, C4) = -1$	→ defocusing, charge scattered
5-sided pyramid: $\chi(T_{1g}, C5) = +\phi$	→ focusing, charge directed to apex

The sign flip between 4-sided (negative) and 5-sided (positive) is the strongest prediction: a 4-sided pyramid at the same slope should produce LESS apex charge than the uncharged baseline, not merely less than 5-sided.

Predicted apex charge ratio:

$$5\text{-sided} / 4\text{-sided} = |\chi(C5)| / |\chi(C4)| = \phi / 1 = \phi = 1.618 \quad [\text{AC13}]$$

4.3 SCALING PREDICTION

Because the mechanism is driven by face-area cell compression (not edge-line compression), the escalator signal scales as face area, which scales as k^2 for a linear scale factor k . Comparing a 5 cm base pyramid to a 10 cm base pyramid ($k=2$): signal ratio = 4x, not 2x. This is a clean discriminator between a face-area mechanism and an edge-length mechanism. [AC14]

4.4 EXPERIMENTAL PROTOCOL

Build three pyramids in identical conductive metal (aluminium or copper), identical apex sharpness, identical base size, but $N = 3, 4,$ and 5 sides:

- 5-sided: face slope 31.72° , base pentagon
- 4-sided: face slope 31.72° (same slope, square base)
- 3-sided: face slope 31.72° (same slope, triangular base)

Using the same face slope across all three controls for the classical lightning-rod effect (tip sharpness), which would predict equal charge at all three apexes. A measured ratio of 5-sided/4-sided $\approx \phi = 1.618$ and 3-sided ≈ 0 would confirm the mechanism.

Equipment: electrometer ($\sim \$30$ - $\$50$), charged plate to drive current, three pyramids ($\sim \$10$ - $\$20$ in sheet aluminium or brass). Total bench cost $< \$50$.

For the scaling test: build a second set of pyramids at $2\times$ linear size.

If the escalator effect is real, the 10 cm pyramid should produce $4\times$ the apex charge of the 5 cm pyramid at the same driving voltage, not $2\times$.

4.5 AMBIENT ELECTRON SOURCE AND COOLING PREDICTION

The escalator is the same mechanism as piezoelectricity (Section 3), not a different one. In a piezo crystal, the cells are normally misaligned; applied pressure deforms them toward the Maxwell-critical icosahedral angle, temporarily activating the T_{1g} coupling and separating charge. The pentagonal pyramid (see Section 3.2c) is the permanent version: its face slope $\arctan(1/\phi) = 31.72^\circ$ matches the icosahedral face orientation exactly at rest. The T_{1g} coupling is always active; no stress is needed. The escalator runs continuously, aggregating free electrons from the surrounding environment — not from the pyramid's own conduction electrons.

Ambient free electrons come from natural ionization (cosmic rays + natural radioactivity produce $\sim 2 \times 10^6$ ion pairs per m^3 per second at sea level, giving an equilibrium free electron density $\sim 10^9/\text{m}^3$). These are continuously replenished. The T_{1g} gradient guides approaching electrons toward the apex; the compressed cells block lateral exit, exactly as in Section 4.1.

Mode 1 (isolated pyramid): accumulates to Coulomb saturation $Q_{\text{sat}} \sim 10^{-11}\text{ C}$ in < 1 second. Apex stays charged; any leakage is immediately refilled. Passive, self-maintaining, no power input required.

Mode 2 (base wire to ground): continuous current; ground is the electron source. Rate set by $\eta \times \chi$. A stationary 5-sided pyramid near ground should produce a continuous DC current with no moving parts.

COOLING PREDICTION: each collected electron carries thermal kinetic energy $\sim (3/2)kT \approx 0.038\text{ eV}$ from the ambient air. When guided up the T_{1g} pressure gradient this becomes potential energy — the kinetic energy does not return to the air. The surrounding air loses the thermal energy of every collected electron: $P_{\text{cooling}} = (3/2)kT/e \times I \approx 0.038\text{ V} \times I$. Negligible at pA

currents from ambient ionization; measurable if a stronger electron source (plasma, UV ionization) drives higher collection rates.

SECTION 5: RETRODICTIONS — EXISTING EVIDENCE

The following observations are consistent with T_{1g} coupling at icosahedral geometry. None were predicted by torsionverse before being observed; they are retrodictions that the mechanism naturally accounts for:

Chladni patterns (standing wave nodes): sand grains accumulate at pressure nodes of standing acoustic waves. The T_{1g} minimum is the low-pressure point where the medium coupling is weakest. Small objects (lighter than the medium displacement threshold) accumulate here, distinct from larger objects driven by radiation pressure. This explains why Chladni patterns are mass-dependent: large grains follow radiation pressure (outward); small grains follow the T_{1g} minimum (inward to node). [doc_acoustics.txt]

Acoustic levitation: objects levitate at pressure nodes of standing acoustic waves. In the torsionverse picture: the T_{1g} minimum at the node is a stable orbit for objects coupled to the medium pressure mode. Standard acoustic levitation theory (radiation pressure) gives the same qualitative result but does not predict N-dependence of levitation stability for arrays of sources.

Ringling rocks: certain rocks ring metallically when struck, suggesting internal crystal resonance at icosahedral packing frequencies. The T_{1g} mode at these frequencies produces an internal pressure minimum that reduces the effective weight of the rock when driven at resonance. This is the acoustic analog of the spinning pyramid effect (Section 6 of companion paper doc_series2_spinning_pyramid.txt).

Newgrange chamber acoustics: the Newgrange passage tomb chamber has a computed spherical-cavity fundamental of ~57 Hz and a Helmholtz (passage+chamber) mode of ~1.2 Hz. Published archaeoacoustic measurements report the chamber resonance at ~110 Hz. The spherical cavity estimate is within a factor of 2; the discrepancy is consistent with the non-spherical actual chamber geometry. [AC8-AC9, PASS]

Stonehenge inner horseshoe: the five trilithon pairs of the inner horseshoe form a C₅-symmetric arrangement. This places a T_{1g} focus ($\chi = \phi$) at the altar stone position – the geometric and acoustically optimal location within the structure. This is a retrodiction of the altar stone placement, not a prediction.

SECTION 6: DISCUSSION

6.1 THE UNIFYING QUANTITY IS ALWAYS $\chi(T_{1g})$

Piezoelectricity, charge concentration, acoustic node patterns, and ringing rock resonance are all governed by $\chi(T_{1g}, C_N)$ from the I_h character table. The same group-theoretic number $\phi = 1.618$ appears as:

- The T_{1g} coupling maximum at C₅ (72 deg)
- The predicted 5-sided/4-sided charge ratio
- The T_{1g} pressure-mode character that drives piezo activation
- The ϕ^2 A_g enhancement for two-source acoustic interference [AC10]
- The golden angle (137.5 deg = $360 - \phi \cdot 180$) in phyllotaxis
- The N=10 ring coupling $\chi = \phi^2$ [AC5]

This convergence is not coincidental within the torsionverse framework:

ϕ is the character of the I_h T_{1g} mode at the C₅ rotation, and C₅ (72 deg, pentagonal) is the most efficient coupling angle in the icosahedral medium. Any physical system with approximate 5-fold symmetry

will couple preferentially to T_{1g} at this angle.

6.2 CONNECTION TO SERIES 1

The T_{1g} mode is identified in Series 1 as the EM/pressure mode of the icosahedral medium — the same mode responsible for electrostatics and gravity at macroscopic scales (doc_jobson_cell.txt). The electron escalator and piezoelectricity are human-scale manifestations of the same T_{1g} pressure-following behavior that underlies all of electromagnetism in the torsionverse. The medium does not distinguish between nuclear-scale and bench-scale T_{1g} activation; the coupling law $\chi(T_{1g}, C_N)$ is the same at all scales.

6.3 OPEN ITEMS

- Formal derivation of piezo coefficient magnitude from medium shear modulus G_{shear} , density ρ , and R_s (the proximity-to-Maxwell-criticality parameter is not yet quantified for specific crystals).
- Quasicrystal anomalous EM response measurement: Al-Mn or Icosahedrite at exact Maxwell criticality predicted to show anomalous T_{1g} response under vibration; no measurement currently exists.
- Optimal chiral icosahedral material: permanent T_{1g} coupling without applied stress. Design target for zero-stress voltage generation.

SECTION 7: CONCLUSIONS

Two classes of T_{1g} coupling have been derived:

1. Piezoelectricity: stress drives a crystal toward Maxwell criticality

($3V-E=6$), activating T_{1g}. Voltage $\propto \chi(T_{1g}, C_N)$. Optimal stress axis predicts a-axis > c-axis for quartz. Quasicrystals predicted to show anomalous EM response at icosahedral resonance frequencies.

2. Electron escalator: pentagonal pyramid at face slope $\arctan(1/\phi) =$

31.72 deg presents the icosahedral face orientation on all five faces. Cell compression channels charge to apex. Predicted ratio: 5-sided / 4-sided = $\phi = 1.618$ [AC13]. Effect scales as face area k^2 [AC14], not edge length k . 4-sided defocuses; 3-sided is null.

Both predictions are falsifiable at bench scale for under \$50. The

unifying parameter throughout is $\chi(T_{1g}, C_5) = \phi$ from the I_h

character table. [14/14 PASS, AC1-AC14]

CONTENT SOURCES

- docs/doc_acoustics.txt: AC retrodictions, Newgrange, Stonehenge
- analysis/demos/acoustics_doc.py: AC1-AC14, 14/14 PASS
- experiments/bench_pyramid_escalator.txt: construction protocol
- experiments/bench_pyramid_acoustic.txt: acoustic escalator protocol
- docs/doc_series2_synthesis.txt: Series 1 DOI's for back-references
- docs/series2/doc_acoustics.txt: companion acoustics paper